

Guideline Title: Needle/Simple Thoracostomy

Guideline Number: GUID-3203-PR017

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 12/10/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To optimize oxygenation, prevent/reverse hypoxia in cases of tension pneumothorax, and to relieve intra-thoracic pressure being placed on the mediastinum causing hemodynamic compromise in cases of tension pneumothorax.
- II. **DEFINITIONS:** None
- III. **DEPARTMENT GUIDELINE:**
 - a. **EQUIPMENT:**
 - i. Cleaning solution: Alcohol, Betadine, or chlorohexidine.
 - ii. Needle
 1. 10-14 gauge, ≥ 3-inch catheter over the needle device for adult patients.
 2. 14-18-gauge 2-inch catheter over the needle device for pediatric patients.
 - iii. Simple
 1. Sterile Gloves, 11-blade scalpel, curved forceps, Tegaderm dressing
 - b. **INDICATIONS:** Signs of tension pneumothorax include: Increased HR, RR, work of breathing, PIP and MAP in intubated patients with positive pressure ventilation. Decreased SPO2, lung sounds to the affected side(s), chest rise on affected side(s). JVD in euvoletic patient, subcutaneous emphysema, hypotension, tracheal deviation (late sign), hyper-resonance upon percussion, eventual cardiac arrest.
 - c. **CONTRAINDICATIONS:** ***Simple thoracostomy may only be performed in the setting of traumatic cardiac arrest with suspicion of tension pneumothorax, in all other scenarios needle thoracostomy should be performed***
 - d. **PROCEDURE Needle thoracostomy:**
 - i. Assess airway/breathing/circulation, & clinical signs/symptoms of tension pneumothorax.
 - ii. Administer high concentration oxygen.
 - iii. Monitor the patient's vital signs HR, ECG, RR, SPO2, BP, and ETCO2 in intubated patients.
 - iv. Identify the site: Locate the 2nd intercostal space in the mid-clavicular line or the 4th or 5th intercostal space in the mid-anterior-axillary line of the affected side.
 - v. Prep this site with cleaning solution.
 - vi. Insert the decompression needle into the site and burry the needle to the hub.

Guideline Title: Needle/Simple Thoracostomy

Guideline Number: GUID-3203-PR017

- vii. Remove the needle leaving the catheter in place and listen for release of air.
- viii. Assess for desired effect and complications. Assess lung sounds. Continue to monitor vital signs.
- ix. Continually reassess the patient for signs and symptoms of a redeveloping tension pneumothorax. If signs and symptoms re-develop, additional needle thoracostomies may be indicated.
- e. **PROCEDURE Simple thoracostomy:**
 - i. Utilize only the setting of adult (>16 yo) traumatic cardiac arrest with suspicion for tension pneumothorax
 - ii. Identify the site: Locate the 4th or 5th intercostal space at the midaxillary line (usually above the nipple line, lift breast tissue out of the way if necessary). When in doubt, use a more cephalad (toward the head) approach to avoid inadvertently entering the peritoneal space.
 - iii. Prep this site with cleaning solution.
 - iv. Using scalpel, make a 1-2 inch incision over the rib. The scalpel is used for skin incision ONLY
 - v. Use the curved forceps for blunt dissection, and then insert them through the intercostal muscles into the pleural space.
 - vi. Open the forceps to allow expulsion of air and blood, then remove them. A finger may be gently inserted to confirm entrance into the pleural space
 - vii. Use occlusive dressing to cover wound
 - viii. This procedure may be performed bilaterally if ROSC is not achieved after initial procedure
- IV. **DOCUMENTATION:** EMS Charts
- V. **REFERENCES:** None